

Dickman Approximation of Randomly Weighted Sums via Stein's Method

Chinmoy Bhattacharjee

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The Dickman Function

Smooth Numbers (Dickman 1930)

For $n \in \mathbb{N}$, let ξ_n be uniform on $[n] := \{1, \dots, n\}$ having the prime factorisation $\xi_n = p_1(\xi_n) \cdot \dots \cdot p_r(\xi_n)$ with $p_1(\xi_n) \geq \dots \geq p_r(\xi_n)$. Then for any $a > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(p_1(\xi_n) \leq n^{1/a}) = \rho(a).$$

The **Dickman function** $\rho : [0, \infty) \rightarrow [0, \infty)$ solves the delay differential equation

$$x\rho'(x) + \rho(x-1) = 0, \quad x \in (1, \infty),$$

with the initial condition $\rho(x) = 1$ for all $x \in [0, 1]$.

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- A random variable X on $[0, \infty)$ is distributed as standard Dickman if and only if

$$X \stackrel{d}{=} U(1 + X),$$

where $U \sim \mathbb{U}[0, 1]$ is independent of X .

Generalized Dickman Distributions

➤ More generally, for $\theta > 0$ and i.i.d. $U_1, U_2, \dots \sim \mathbb{U}[0, 1]$, the distribution of

$$D_\theta = U_1^{1/\theta} + U_1^{1/\theta} U_2^{1/\theta} + U_1^{1/\theta} U_2^{1/\theta} U_3^{1/\theta} + \dots$$

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- Often appear as limits of weighted random sums; applications in number theory, random graphs and so on.

Random Sums

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- (Smooth Numbers) Let $p_1 < p_2 < \dots$ be an enumeration of primes. Define

$$\Omega_n := \{N \in \mathbb{N} : \text{for } p \text{ prime, } p \mid N \implies p \leq p_n\}.$$

Let $M_n = \prod_{k=1}^n p_k^{X_k}$ be a random variable on Ω_n with $\mathbb{P}(M_n = N) \propto \frac{1}{N}$.

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$$\frac{\log M_n}{\log p_n} = \frac{1}{\log p_n} \sum_{k=1}^n \log p_k X_k \xrightarrow{d} D \quad \text{as } n \rightarrow \infty. \quad (\text{Pinsky '19})$$

Rates of Convergence

For non-negative random variables X, Y , define

Kolmogorov distance: $d_K(X, Y) = \sup_{t \geq 0} |\mathbb{P}(X \leq t) - \mathbb{P}(Y \leq t)|$

Smooth distance: $d_{1,1}(X, Y) = \sup_{h, h' \in \text{Lip}(1)} |\mathbb{E}h(X) - \mathbb{E}h(Y)|$

Theorem (Bhattacharjee/Goldstein '19)

Let $B_k \sim \text{Ber}(\frac{1}{k})$ be independent and $W_n := \frac{1}{n} \sum_{k=1}^n kB_k$. Then,

$$d_{1,1}(W_n, D) \leq \frac{3}{4n}.$$

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Theorem (Bhattacharjee/Schulte '25)

Fix $\theta > 0$, $\beta \in \mathbb{R}$. Let $W_n := \frac{1}{n} \sum_{k=1}^n kB_k$ for $n \geq l \geq \theta - \beta$, where $B_k \sim \text{Ber}(\frac{\theta}{k+\beta})$ are independent. Then

$$d_K(W_n, D_\theta) \leq \begin{cases} \frac{C(l + |\beta| \log(n/l))}{n}, & \theta \geq 1, \\ \frac{Cl^\theta}{n^\theta}, & \theta \in (0, 1). \end{cases}$$

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Motivation: Can we obtain similar bounds on d_K for more general weighted sums?

A general framework

Consider the randomly weighted sum $S_n = \sum_{k=1}^n W_{n,k} X_k$, $n \in \mathbb{N}$.

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(A1) Ordered Weights: Let $\mathbf{W} = (W_{n,k})_{n \in \mathbb{N}, k \in [n]}$ be an array of ordered non-negative random variables

$$0 := W_{n,0} \leq W_{n,1} \leq \cdots \leq W_{n,n} := W_{\max}^{(n)},$$

with $\mathbb{E}[W_{\max}^{(n)}] < \infty$.

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- We do **not** assume independence among the weights $\mathbf{W} = (W_{n,k})$; can be highly dependent (e.g., order statistics).
- X_k may depend on $\mathbf{W}$. We only require $(X_k)_{k \in \mathbb{N}}$ to be conditionally independent.

Main Abstract Bound

Theorem (Bhattacharjee '26+)

Let $S_n = \sum_{k=1}^n W_{n,k} X_k$ satisfy the ordering and conditional independence assumptions. Then, there exist $C_1, C_2 \in (0, \infty)$ depending only on θ such that

$$d_K(S_n, D_\theta) \leq \begin{cases} C_1 \mathbb{E}[\varepsilon_n(\mathbf{W})], & \text{if } \theta \geq 1, \\ C_2 (\mathbb{E}[\varepsilon_n(\mathbf{W})] + \mathbb{E}[\varepsilon_n(\mathbf{W})]^\theta), & \text{if } \theta \in (0, 1). \end{cases}$$

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- Here $\varepsilon_n(\mathbf{W})$ is the aggregate error term quantifying certain discretization errors and conditional Poisson approximation of the counts (explicit bounds on the next slide).
- Building upon the Stein's method framework of [Bhattacharjee/Schulte '25](#), this significantly generalizes their results by incorporating random weights and dependence.

Anatomy of the Error Term

$$d_K(S_n, D_\theta) \leq \begin{cases} C_1 \mathbb{E}[\varepsilon_n(\mathbf{W})], & \text{if } \theta \geq 1, \\ C_2 (\mathbb{E}[\varepsilon_n(\mathbf{W})] + \mathbb{E}[\varepsilon_n(\mathbf{W})]^\theta), & \text{if } \theta \in (0, 1). \end{cases}$$

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$$\begin{aligned} \varepsilon_n(\mathbf{W}) := & \underbrace{\max_k (W_{n,k} - W_{n,k-1})}_{\text{Maximum step size}} + \underbrace{\frac{2}{\theta} \max_k \left| \sum_{j=1}^k (\mathbb{E}[X_j | \mathbf{W}] W_{n,j} - \theta(W_{n,j} - W_{n,j-1})) \right|}_{\text{Maximum cumulative deviation}} \\ & + \underbrace{\frac{2}{\theta} \sum_{k=1}^n \frac{\mathbb{E}[X_k | \mathbf{W}] W_{n,k} d_{TV}(X_k^*, X_k + 1 | \mathbf{W})}{\mathbb{P}(X_k = x_{\min}^{(k)}(\mathbf{W}) | \mathbf{W})}}_{\text{Poisson approximation of } X_k | \mathbf{W}} + \underbrace{|W_{\max}^{(n)} - 1|}_{\text{Boundary discrepancy}}. \end{aligned}$$

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Here X_k^* denotes the **size-biased version** of $X_k | \mathbf{W}$, and $x_{\min}^{(k)}(\mathbf{W})$ is the smallest integer in its support.

Dickman Limit in Harmonic Measures: Divisor Function

Let $p_1 < p_2 < \cdots$ be an enumeration of primes and

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The Generalized Divisor Function:

For $\theta > 0$, the generalized divisor function $\tau_\theta: \mathbb{N} \rightarrow \mathbb{R}$ is the multiplicative arithmetic function defined on prime powers by

$$\tau_\theta(p^j) := \binom{j+\theta-1}{j} = \frac{\theta(\theta+1)\cdots(\theta+j-1)}{j!}, \quad j \in \mathbb{N}_0.$$

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- **Combinatorial meaning:** When $\theta = k \in \mathbb{N}$, $\tau_k(N)$ exactly counts the number of ways N can be expressed as an ordered product of k integers.
- **Standard harmonic case:** For $\theta = 1$, the function collapses to $\tau_1(N) \equiv 1$.

Harmonic Measures: Dickman Limit

Generalized Harmonic Measure: We endow Ω_n with a the probability measure

$$\mathbb{P}_n(\{N\}) \propto \frac{\tau_\theta(N)}{N}, \quad N \in \Omega_n.$$

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Theorem (Bhattacharjee '26+)

Let $M_n \sim \mathbb{P}_n$. For any $\theta > 0$, there exists constants $0 < C \leq C' < \infty$ such that

$$\frac{C}{(\log n)^{\min\{1, \theta\}}} \leq d_K \left(\frac{\log M_n}{\log p_n}, D_\theta \right) \leq \frac{C'}{(\log n)^{\min\{1, \theta\}}}.$$

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- The same optimal bound holds if we restrict $\mathbb{P}_n$ to the square-free space (non-quantitative result for $\theta = 1$ in [Cellarosi/Sinai '13](#))

$$\Omega_n^{\text{sq}} = \{N \in \mathbb{N} : \text{for } p \text{ prime, } p \mid N \implies p \leq p_n \text{ and } p^2 \nmid N\}.$$

Network Routing (Dependent Increments)

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- **Locations (Weights):** Transmitter locations are order statistics of uniforms. By Rényi's representation, we set $W_{n,k} := \frac{1}{S_n^Y} \sum_{i=1}^k Y_i$ where $Y_i \stackrel{iid}{\sim} \text{Exp}(1)$, $S_n^Y = \sum_{i=1}^n Y_i$.

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- **Traffic (Counts) and Total Cost:** Greedy routing assigns the heaviest traffic to the closest nodes. The k -th closest node generates $X_k \sim \text{Poi}(\theta/k)$ data packets. Total transmission cost is exactly $S_n = \sum W_{n,k} X_k$.

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Theorem (Bhattacharjee '26+)

Assume that $(Y_i)_{i \in \mathbb{N}}$ are independent and non-negative with $\mathbb{E}[Y_i] = 1$ and $\max_{i \in \mathbb{N}} \text{Var}(Y_i) < \infty$. Then

$$d_K(S_n, D_\theta) \leq C \left(\frac{1}{\sqrt{n}} \right)^{\min\{1, \theta\}}.$$

Geometric Sums on CUE Spectra: The Weights

CUE Eigenphases and Mutual Repulsion:

The eigenphases $0 \leq \theta_1 < \cdots < \theta_n < 2\pi$ of an $n \times n$ Circular Unitary Ensemble (CUE) matrix form a Determinantal Point Process (DPP). Their joint density is proportional to

$$\prod_{1 \leq j < k \leq n} |e^{i\theta_j} - e^{i\theta_k}|^2.$$

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Ordered Weights and Rigidity:

Rotating the spectrum by a uniform eigenvalue to set the “ground state” at angle 0, we define the weights via the remaining ordered eigenphases $0 < \phi_1 < \cdots < \phi_{n-1} < 2\pi$:

$$W_{n,k} := \frac{\phi_k}{2\pi}, \quad k \in [n-1].$$

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By rotation invariance and repulsion, the gaps exhibit *rigidity*: $\exists C \in (0, \infty)$ such that

$$\mathbb{E} \left[\max_{k \in [n-1]} \left| W_{n,k} - \frac{k}{n} \right| \right] \leq C \frac{\log n}{n}.$$

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Conditionally on the weights $W_{n,k} := \frac{\phi_k}{2\pi}$, $k \in [n-1]$, we define the sequence $(X_k)_{k \in [n-1]}$ as independent Geometric random variables with mean

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For $S_n = \sum_{k=1}^{n-1} W_{n,k} X_k$, there exist constants $K_1, K_2 \in (0, \infty)$ depending only on θ such that for $n \geq 2$,

$$K_1 \left(\frac{1}{n} \right)^{\min\{1, \theta\}} \leq d_K(S_n, D_\theta) \leq K_2 \left(\frac{\log n}{n} \right)^{\min\{1, \theta\}}.$$

Idea of the Proof I: Stein Equation

Proposition (Goldstein '17)

For $\theta > 0$, a non-negative random variable $W \sim D_\theta$ if and only if

$$\mathbb{E} \left[\frac{W}{\theta} f'(W) - f(W+1) + f(W) \right] = 0$$

for all Lipschitz functions $f : [0, \infty) \rightarrow \mathbb{R}$.

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for all Lipschitz functions $f : [0, \infty) \rightarrow \mathbb{R}$.

For $\theta > 0$ and $a \geq 0$, we consider the Stein equation

$$\frac{x}{\theta} f'(x) + f(x) - f(x+1) = \mathbb{1}(x \leq a) - \mathbb{1}(D_\theta \leq a), \quad x \in [0, \infty).$$

For a solution f_a and an independent $U \sim \text{Uniform}([0, 1])$,

$$\mathbb{P}(S_n \leq a \mid \mathbf{W}) - \mathbb{P}(D_\theta \leq a) = \mathbb{E} \left[\frac{S_n}{\theta} f'_a(S_n) - f'_a(S_n + U) \mid \mathbf{W} \right].$$

Idea of the Proof II: The Size-Biasing and Coupling

Using conditional size-biasing, we have

$$\mathbb{E}[S_n f'_a(S_n) \mid \mathbf{w}] = \sum_{k=1}^n \mathbb{E}(X_k \mid \mathbf{w}) \mathbb{E} \left[W_{n,k} f'_a(S_n^{(k)} + W_{n,k} X_k^*) \mid \mathbf{w} \right]$$

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Idea of the Proof III: The Stein Solution

- (Bhattacharjee/Schulte '25) Let $\theta > 0$. For $a \geq 0$, there exists a solution f_a to the Stein equation, such that a.e.,

$$f'_a = (g_1 + u_+) - (g_2 + u_-),$$

where

- $0 \leq g_1(x) \leq g_2(x) = \frac{\theta}{a} \mathbb{1}(x \leq a) + \frac{\theta a^\theta}{x^{\theta+1}} \mathbb{1}(x > a)$,
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- This yields

$$|\mathbb{P}(S_n \leq a \mid \mathbf{W}) - \mathbb{P}(D_\theta \leq a)| \leq \varepsilon_n(\mathbf{W}) \mathbb{E} \left[\theta^2 + g_2(S_n) \mid \mathbf{W} \right].$$

Idea of the Proof IV: Excluding Small a

➤ (Bhattacharjee/Schulte '25) For $\theta > 0$, $a_0 > 0$ and a random variable $W \geq 0$,

$$d_K(W, D_\theta) \leq \sup_{a \geq a_0} |\mathbb{P}(W \leq a) - \mathbb{P}(D_\theta \leq a)| + R_\theta(a_0)a_0$$

with $R_\theta(s) = s^{\theta-1}$ for $\theta \in (0, 1)$ and $R_\theta(s) = K_\theta \in (0, \infty)$ for $\theta \geq 1$.

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- Pick $a_0 \approx \varepsilon_n(\mathbf{W})$. For $a \geq a_0$, we use

$$|\mathbb{P}(S_n \leq a \mid \mathbf{W}) - \mathbb{P}(D_\theta \leq a)| \leq \varepsilon_n(\mathbf{W}) \mathbb{E}[\theta^2 + g_2(S_n) \mid \mathbf{W}].$$

This yields a bound on $d_K(S_n, D_\theta \mid \mathbf{W})$. Taking a unconditional expectation yields the result.

Further Directions

- The condition of ordered weights can be removed and artificially satisfied by ordering the weights after conditioning. This however seems to provide suboptimal bounds (compared to known $d_{1,1}$ bounds) in the case where the weights are independent with mean k/n .

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Thank you for your attention!